

# torch.std#

<https://pytorch.org/docs/stable/generated/torch.std.html#torch.std>

## Description:

Calculates the standard deviation over the dimensions specified by dim. dim can be a single dimension, list of dimensions, or None to reduce over all dimensions. The standard deviation ( $\sigma$ ) is calculated as where  $x$  is the sample set of elements,  $\bar{x}$  is the sample mean,  $N$  is the number of samples and  $\delta N$  is the correction. If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s). input (Tensor) – the input tensor.

## Function Signature

```
torch.std(input, dim=None, *, correction=1, keepdim=False, out=None) → Tensor#
```

Parameters

Keyword Arguments

## Parameters

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### Keyword

correction (int) – difference between the sample size and sample degrees of freedom. Defaults to Bessel's correction, correction=1. Changed in version 2.0: Previously this argument was called unbiased and was a boolean with True corresponding to correction=1 and False being correction=0. keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False. out (Tensor, optional) – the output tensor.

## Code Examples

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```
>>> a = torch.tensor(
...     [[ 0.2035,  1.2959,  1.8101, -0.4644],
...      [ 1.5027, -0.3270,  0.5905,  0.6538],
...      [-1.5745,  1.3330, -0.5596, -0.6548],
...      [ 0.1264, -0.5080,  1.6420,  0.1992]]
... ) # fmt: skip
>>> torch.std(a, dim=1, keepdim=True)
tensor([[1.0311],
        [0.7477],
        [1.2204],
        [0.9087]])
```

# Detailed Documentation

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## Documentation

Calculates the standard deviation over the dimensions specified by `dim`. `dim` can be a single dimension, list of dimensions, or `None` to reduce over all dimensions.

The standard deviation ( $\sigma$ ) is calculated as

where  $x$  is the sample set of elements,  $\bar{x}$  is the sample mean,  $N$  is the number of samples and  $\frac{1}{N}$  is the correction.

If `keepdim` is `True`, the output tensor is of the same size as input except in the dimension(s) `dim` where it is of size 1. Otherwise, `dim` is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s).

`input (Tensor)` – the input tensor.

`dim (int or tuple of ints, optional)` – the dimension or dimensions to reduce. If `None`, all dimensions are reduced.

difference between the sample size and sample degrees of freedom. Defaults to Bessel's correction, `correction=1`.

Changed in version 2.0: Previously this argument was called `unbiased` and was a boolean with `True` corresponding to `correction=1` and `False` being `correction=0`.

`keepdim (bool, optional)` – whether the output tensor has `dim` retained or not. Default: `False`.

`out (Tensor, optional)` – the output tensor.

